

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# VECTORS, (MULTIPLYING VECTORS BY A SCALAR)

**LO: I can multiply a vector by a scalar and understand its geometric effect.**

## Scalar multiplication of vectors

Multiplying a vector by a **scalar** (number) multiplies each component by that number. This **scales** the vector: changing its magnitude but not its direction (unless negative).

**Key fact:** A negative scalar reverses the direction. Vectors that are scalar multiples of each other are parallel.

## Quick examples

|   |                                    |                                    |
|---|------------------------------------|------------------------------------|
| ● | $3(2, 4) = (6, 12)$                | multiply each component by 3       |
| ● | $-2(1, 3) = (-2, -6)$              | negative scalar reverses direction |
| ● | $(4, 6)$ and $(2, 3)$ are parallel | one is 2 times the other           |
| ● | $3(2, 4) = (6, 4)$                 | must multiply both components      |

## Key idea

Multiply each component by the scalar. Parallel vectors are scalar multiples of each other.

$$k(a, b) = (ka, kb)$$

Scales magnitude by  $|k|$ , reverses direction if  $k < 0$

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 – positive scalar

Calculate  $4(2, 3)$ .

$$\text{Multiply } x: 4 \times 2 = 8$$

$$\text{Multiply } y: 4 \times 3 = 12$$

$$\text{Result: } (8, 12)$$

$$(8, 12)$$

Each component is multiplied by the scalar.

### Example 2 – negative scalar

Calculate  $-3(1, -2)$ .

$$\text{Multiply } x: -3 \times 1 = -3$$

$$\text{Multiply } y: -3 \times (-2) = 6$$

$$\text{Result: } (-3, 6)$$

$$(-3, 6)$$

Negative scalar reverses direction.

### Example 3 – showing vectors are parallel

Show that  $(6, 9)$  and  $(2, 3)$  are parallel.

Check if one is a scalar multiple of the other

$$(6, 9) = 3(2, 3)$$

Since  $(6, 9) = 3 \times (2, 3)$ , they are parallel

**Parallel (scalar factor 3)**

If  $a = kb$  for some scalar  $k$ , the vectors are parallel.

### Example 4 – combining scalar and addition

If  $a = (3, 1)$  and  $b = (2, -4)$ , find  $2a - 3b$ .

$$2a = 2(3, 1) = (6, 2)$$

$$3b = 3(2, -4) = (6, -12)$$

$$2a - 3b = (6, 2) - (6, -12) = (0, 14)$$

$$(0, 14)$$

Scalar multiply first, then subtract.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Positive scalars

Calculate each scalar multiple.

- a)  $2(3, 5)$
- b)  $5(1, -2)$
- c)  $3(-4, 2)$
- d)  $4(0, 7)$

### 2. Negative scalars

Calculate each.

- a)  $-2(4, 1)$
- b)  $-1(5, -3)$
- c)  $-3(2, 6)$
- d)  $-4(-1, 2)$

### 3. Fractional scalars

Calculate each.

- a)  $\frac{1}{2}(6, 4)$
- b)  $\frac{1}{3}(9, -6)$
- c)  $\frac{2}{3}(12, 9)$
- d)  $-\frac{1}{2}(8, -10)$

### 4. Parallel vectors

State whether the vectors are parallel. If so, give the scalar.

- a)  $(4, 6)$  and  $(2, 3)$
- b)  $(3, 5)$  and  $(6, 10)$
- c)  $(1, 2)$  and  $(3, 5)$
- d)  $(-4, 8)$  and  $(2, -4)$

### 5. Combined operations

Given  $a = (2, 3)$  and  $b = (4, -1)$ , calculate:

- a)  $3a$
- b)  $a + 2b$
- c)  $3a - b$
- d)  $2a + 3b$
- e)  $-a + 4b$

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a)  $3(2, 4) = (6, 12)$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b)  $3(2, 4) = (6, 4)$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) Parallel vectors are scalar multiples of each other ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d)  $-1(3, 2) = (3, 2)$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) If  $a = 2b$ , then  $a$  and  $b$  are parallel and  $a$  is twice as long ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Vectors  $p = (2, 5)$  and  $q = (k, 15)$  are parallel. Find the value of  $k$ . Write  $q$  as a scalar multiple of  $p$ .

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Points A, B, C have position vectors  $a = (1, 2)$ ,  $b = (4, 5)$ ,  $c = (10, 11)$ . Show that A, B, and C are collinear by showing that AB is a scalar multiple of AC.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

